1) Create a class Student with instance fields name, age and a class variable count to track number of students. Use local variables to assign values and display details.

```
class Student {
    // Instance variables
    String name;
    int age;

    // Class variable (static)
    static int count = 0;

    // Constructor
    Student(String name, int age) {
        this.name = name;
        this.age = age;
        count++;
    }

    // Display student details
    void display() {
        // Local variable
        String college = "COER University";

        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("College: " + college);
        System.out.println();
    }

    public static void main(String[] args) {

        // Local variables
        String studentName1 = "Sumit";
        int studentAge1 = 20;

        String studentName2 = "Rahul";
        int studentAge2 = 21;

        // Creating objects
        Student s1 = new Student(studentName1, studentAge1);
        Student s2 = new Student(studentName2, studentAge2);

        // Display details
```

```
        s1.display();
        s2.display();

        System.out.println("Total Students: " + Student.count);
    }
}
```

OUTPUT:
Name: Sumit
Age: 20
College: COER University

Name: Rahul
Age: 21
College: COER University

Total Students: 2

2) Create a class Rectangle with instance fields length, breadth, a class variable shapeName = "Rectangle". Write methods with local variables to calculate and display the area.

```
class Rectangle {
    // Instance variables
    double length;
    double breadth;

    // Class variable
    static String shapeName = "Rectangle";

    // Constructor
    Rectangle(double length, double breadth) {
        this.length = length;
        this.breadth = breadth;
    }

    // Method to calculate and display area
    void displayArea() {
        // Local variable
        double area = length * breadth;

        System.out.println("Shape Name: " + shapeName);
```

```
        System.out.println("Length: " + length);
        System.out.println("Breadth: " + breadth);
        System.out.println("Area: " + area);
    }

    public static void main(String[] args) {
        Rectangle r1 = new Rectangle(10, 5);

        r1.displayArea();
    }
}
```

OUTPUT:
Shape Name: Rectangle
Length: 10.0
Breadth: 5.0
Area: 50.0

3) Create a class Circle with instance field radius, a class variable PI = 3.14. Use local variables to calculate area and circumference.

```
class Circle {
    // Instance variable
    double radius;

    // Class variable
    static double PI = 3.14;

    // Constructor
    Circle(double radius) {
        this.radius = radius;
    }

    // Method to calculate and display area and circumference
    void display() {
        // Local variables
        double area = PI * radius * radius;
        double circumference = 2 * PI * radius;

        System.out.println("Radius: " + radius);
        System.out.println("Area of Circle: " + area);
        System.out.println("Circumference of Circle: " + circumference);
    }
```

```
    public static void main(String[] args) {
        Circle c1 = new Circle(5);

        c1.display();
    }
}
```

OUTPUT:
Radius: 5.0
Area of Circle: 78.5
Circumference of Circle: 31.400000000000002

4) Create a class Employee with instance fields empId, salary, and a class variable companyName. Display employee details using local variables.

```
class Employee {
    // Instance variables
    int empId;
    double salary;

    // Class variable
    static String companyName = "ABC Company";

    // Constructor
    Employee(int empId, double salary) {
        this.empId = empId;
        this.salary = salary;
    }

    // Method to display employee details
    void displayDetails() {
        // Local variable
        String message = "Employee Details";

        System.out.println(message);
        System.out.println("Employee ID: " + empId);
        System.out.println("Salary: " + salary);
        System.out.println("Company Name: " + companyName);
    }

    public static void main(String[] args) {
        // Local variables
        int id = 101;
```

```
        double empSalary = 50000;

        Employee e1 = new Employee(id, empSalary);

        e1.displayDetails();
    }
}
```

OUTPUT:
Employee Details
Employee ID: 101
Salary: 50000.0
Company Name: ABC Company

5) Create a class Book with fields title, author, price, and a class variable publisher. Use local variables to set and display book details.

```
class Book {
    // Instance variables
    String title;
    String author;
    double price;

    // Class variable
    static String publisher = "ABC Publishers";

    // Constructor
    Book(String title, String author, double price) {
        this.title = title;
        this.author = author;
        this.price = price;
    }

    // Method to display book details
    void displayDetails() {
        // Local variable
        String message = "Book Details";

        System.out.println(message);
        System.out.println("Title: " + title);
        System.out.println("Author: " + author);
        System.out.println("Price: " + price);
        System.out.println("Publisher: " + publisher);
```

```
    }

    public static void main(String[] args) {

        // Local variables
        String bookTitle = "Java Programming";
        String bookAuthor = "James";
        double bookPrice = 450.0;

        // Create object
        Book b1 = new Book(bookTitle, bookAuthor, bookPrice);

        // Display details
        b1.displayDetails();
    }
}
```

OUTPUT:
Book Details
Title: Java Programming
Author: James
Price: 450.0
Publisher: ABC Publishers

6) Create a class Car with fields brand, mileage, and a class variable wheels = 4. Display details using local variables.

```
class Car {
    // Instance variables
    String brand;
    double mileage;

    // Class variable
    static int wheels = 4;

    // Constructor
    Car(String brand, double mileage) {
        this.brand = brand;
        this.mileage = mileage;
    }
```

```
    // Method to display car details
    void displayDetails() {
        // Local variable
        String message = "Car Details";

        System.out.println(message);
        System.out.println("Brand: " + brand);
        System.out.println("Mileage: " + mileage + " km/l");
        System.out.println("Number of Wheels: " + wheels);
    }

    public static void main(String[] args) {

        // Local variables
        String carBrand = "Honda";
        double carMileage = 18.5;

        // Create object
        Car c1 = new Car(carBrand, carMileage);

        // Display details
        c1.displayDetails();
    }
}
```

OUTPUT:
Car Details
Brand: Honda
Mileage: 18.5 km/l
Number of Wheels: 4

7) Create a class BankAccount with fields accountNumber, balance, and a class variable bankName. Use local variables in deposit method.

```
class BankAccount {
    // Instance variables
    int accountNumber;
    double balance;

    // Class variable
    static String bankName = "State Bank of India";

    // Constructor
```

```
    BankAccount(int accountNumber, double balance) {
        this.accountNumber = accountNumber;
        this.balance = balance;
    }

    // Deposit method
    void deposit(double amount) {
        // Local variable
        double newBalance = balance + amount;

        balance = newBalance;

        System.out.println("Amount Deposited: " + amount);
        System.out.println("New Balance: " + newBalance);
    }

    // Display account details
    void displayDetails() {
        System.out.println("Bank Name: " + bankName);
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    public static void main(String[] args) {
        BankAccount account = new BankAccount(123456, 5000);

        account.displayDetails();

        account.deposit(2000);
    }
}
```

OUTPUT:
Bank Name: State Bank of India
Account Number: 123456
Balance: 5000.0

Amount Deposited: 2000.0
New Balance: 7000.0

8) Create a class Pen with fields color, type, and a class variable manufacturer. Use local variables to display pen details.

```
class Pen {
```

```
    // Instance variables
    String color;
    String type;

    // Class variable
    static String manufacturer = "Cello";

    // Constructor
    Pen(String color, String type) {
        this.color = color;
        this.type = type;
    }

    // Method to display pen details
    void displayDetails() {
        // Local variable
        String message = "Pen Details";

        System.out.println(message);
        System.out.println("Color: " + color);
        System.out.println("Type: " + type);
        System.out.println("Manufacturer: " + manufacturer);
    }

    public static void main(String[] args) {

        // Local variables
        String penColor = "Blue";
        String penType = "Ball Pen";

        // Create object
        Pen p1 = new Pen(penColor, penType);

        // Display details
        p1.displayDetails();
    }
}
```

OUTPUT:
Pen Details
Color: Blue
Type: Ball Pen
Manufacturer: Cello

9) Create a class Laptop with fields brand, RAM, and a class variable os = "Windows". Use local variables to display details.

```
class Laptop {
    // Instance variables
    String brand;
    int RAM;

    // Class variable
    static String os = "Windows";

    // Constructor
    Laptop(String brand, int RAM) {
        this.brand = brand;
        this.RAM = RAM;
    }

    // Method to display laptop details
    void displayDetails() {
        // Local variable
        String message = "Laptop Details";

        System.out.println(message);
        System.out.println("Brand: " + brand);
        System.out.println("RAM: " + RAM + " GB");
        System.out.println("Operating System: " + os);
    }

    public static void main(String[] args) {

        // Local variables
        String laptopBrand = "HP";
        int laptopRAM = 16;

        // Create object
        Laptop l1 = new Laptop(laptopBrand, laptopRAM);

        // Display details
        l1.displayDetails();
    }
}
```

OUTPUT:

Laptop Details
Brand: HP
RAM: 16 GB
Operating System: Windows

10) Create a class Movie with fields name, genre, rating, and a class variable industry. Display movie details using local variables.

```
class Movie {
    // Instance variables
    String name;
    String genre;
    double rating;

    // Class variable
    static String industry = "Bollywood";

    // Constructor
    Movie(String name, String genre, double rating) {
        this.name = name;
        this.genre = genre;
        this.rating = rating;
    }

    // Method to display movie details
    void displayDetails() {
        // Local variable
        String message = "Movie Details";

        System.out.println(message);
        System.out.println("Name: " + name);
        System.out.println("Genre: " + genre);
        System.out.println("Rating: " + rating);
        System.out.println("Industry: " + industry);
    }

    public static void main(String[] args) {

        // Local variables
        String movieName = "3 Idiots";
        String movieGenre = "Comedy/Drama";
        double movieRating = 8.4;
```

```
        // Create object
        Movie m1 = new Movie(movieName, movieGenre, movieRating);

        // Display details
        m1.displayDetails();
    }
}
```

OUTPUT:
Movie Details
Name: 3 Idiots
Genre: Comedy/Drama
Rating: 8.4
Industry: Bollywood

Intermediate Level (Q11–Q20)
Focus: encapsulation, conditions, multiple methods, local variables, and class variables.

11) Create a class Calculator with methods for add, subtract, multiply, divide using local variables. Add a class variable operationsCount to count total operations done.

```
class Calculator {

    // Class variable
    static int operationsCount = 0;

    // Addition method
    void add(int a, int b) {
        int result = a + b; // Local variable
        operationsCount++;

        System.out.println("Addition = " + result);
    }

    // Subtraction method
    void subtract(int a, int b) {
        int result = a - b; // Local variable
        operationsCount++;

        System.out.println("Subtraction = " + result);
    }

    // Multiplication method
```

```
    void multiply(int a, int b) {
        int result = a * b; // Local variable
        operationsCount++;

        System.out.println("Multiplication = " + result);
    }

    // Division method
    void divide(int a, int b) {
        if (b != 0) {
            double result = (double) a / b; // Local variable
            operationsCount++;

            System.out.println("Division = " + result);
        } else {
            System.out.println("Cannot divide by zero.");
        }
    }

    public static void main(String[] args) {

        Calculator c1 = new Calculator();

        // Perform operations
        c1.add(10, 5);
        c1.subtract(10, 5);
        c1.multiply(10, 5);
        c1.divide(10, 5);

        System.out.println("Total Operations Performed: "
                + operationsCount);
    }
}
```

OUTPUT:
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2.0
Total Operations Performed: 4

12) Create a class StudentMarks with fields marks1, marks2, marks3, and a class variable

universityName. Use local variables to calculate average.

```
class StudentMarks {

    // Instance variables
    double marks1;
    double marks2;
    double marks3;

    // Class variable
    static String universityName = "COER University";

    // Constructor
    StudentMarks(double marks1, double marks2, double marks3) {
        this.marks1 = marks1;
        this.marks2 = marks2;
        this.marks3 = marks3;
    }

    // Method to calculate and display average
    void displayDetails() {

        // Local variable
        double average = (marks1 + marks2 + marks3) / 3;

        System.out.println("University: " + universityName);
        System.out.println("Marks 1: " + marks1);
        System.out.println("Marks 2: " + marks2);
        System.out.println("Marks 3: " + marks3);
        System.out.println("Average Marks: " + average);
    }

    public static void main(String[] args) {

        StudentMarks s1 = new StudentMarks(80, 75, 90);

        s1.displayDetails();
    }
}
```

OUTPUT:

```
University: COER University
Marks 1: 80.0
Marks 2: 75.0
Marks 3: 90.0
```

Average Marks: 81.66666666666667

13) Create a class Temperature with a field celsius and a class variable scale = "Celsius". Use local variables to convert into Fahrenheit and Kelvin.

```
class Temperature {

    // Instance variable
    double celsius;

    // Class variable
    static String scale = "Celsius";

    // Constructor
    Temperature(double celsius) {
        this.celsius = celsius;
    }

    // Method to convert and display temperature
    void convertTemperature() {

        // Local variables
        double fahrenheit = (celsius * 9 / 5) + 32;
        double kelvin = celsius + 273.15;

        System.out.println("Temperature in " + scale + ": " + celsius);
        System.out.println("Temperature in Fahrenheit: " + fahrenheit);
        System.out.println("Temperature in Kelvin: " + kelvin);
    }

    public static void main(String[] args) {

        Temperature t1 = new Temperature(25);

        t1.convertTemperature();
    }
}
```

OUTPUT:
Temperature in Celsius: 25.0
Temperature in Fahrenheit: 77.0
Temperature in Kelvin: 298.15

14) Create a class Person with fields name, age and a class variable country. Use local variables to check voting eligibility.

```
class Person {

    // Instance variables
    String name;
    int age;

    // Class variable
    static String country = "India";

    // Constructor
    Person(String name, int age) {
        this.name = name;
        this.age = age;
    }

    // Method to check voting eligibility
    void checkVotingEligibility() {

        // Local variable
        String eligibility;

        if (age >= 18) {
            eligibility = "Eligible to vote";
        } else {
            eligibility = "Not eligible to vote";
        }

        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Country: " + country);
        System.out.println("Voting Status: " + eligibility);
    }

    public static void main(String[] args) {

        // Local variables
        String personName = "Sumit";
        int personAge = 20;

        Person p1 = new Person(personName, personAge);
```

```
        p1.checkVotingEligibility();
    }
}
```

OUTPUT:
Name: Sumit
Age: 20
Country: India
Voting Status: Eligible to vote

15) Create a class SimpleInterest with fields principal, rate, time, and a class variable bank. Use local variables to calculate interest.

```
class SimpleInterest {

    // Instance variables
    double principal;
    double rate;
    double time;

    // Class variable
    static String bank = "State Bank of India";

    // Constructor
    SimpleInterest(double principal, double rate, double time) {
        this.principal = principal;
        this.rate = rate;
        this.time = time;
    }

    // Method to calculate interest
    void calculateInterest() {
        // Local variable
        double interest = (principal * rate * time) / 100;

        System.out.println("Bank: " + bank);
        System.out.println("Principal: " + principal);
        System.out.println("Rate: " + rate + "%");
        System.out.println("Time: " + time + " years");
        System.out.println("Simple Interest: " + interest);
    }

    public static void main(String[] args) {
```

```
        // Local variables
        double p = 10000;
        double r = 5;
        double t = 2;

        SimpleInterest s = new SimpleInterest(p, r, t);
        s.calculateInterest();
    }
}
```

OUTPUT:
Bank: State Bank of India
Principal: 10000.0
Rate: 5.0%
Time: 2.0 years
Simple Interest: 1000.0

16) Create a class Triangle with sides a, b, c, and a class variable type = "Scalene/Isosceles/Equilateral". Use local variables to check validity.

```
class Triangle {

    // Instance variables
    double a;
    double b;
    double c;

    // Class variable
    static String type = "Scalene/Isosceles/Equilateral";

    // Constructor
    Triangle(double a, double b, double c) {
        this.a = a;
        this.b = b;
        this.c = c;
    }

    // Method to check triangle validity
    void checkTriangle() {

        // Local variable
        String result;
```

```
        if (a + b > c && a + c > b && b + c > a) {

            if (a == b && b == c) {
                result = "Equilateral Triangle";
            } else if (a == b || b == c || a == c) {
                result = "Isosceles Triangle";
            } else {
                result = "Scalene Triangle";
            }

        } else {
            result = "Invalid Triangle";
        }

        System.out.println("Side a: " + a);
        System.out.println("Side b: " + b);
        System.out.println("Side c: " + c);
        System.out.println("Possible Types: " + type);
        System.out.println("Result: " + result);
    }

    public static void main(String[] args) {

        // Local variables
        double side1 = 5;
        double side2 = 5;
        double side3 = 8;

        Triangle t1 = new Triangle(side1, side2, side3);

        t1.checkTriangle();
    }
}
```

OUTPUT:
Side a: 5.0
Side b: 5.0
Side c: 8.0
Possible Types: Scalene/Isosceles/Equilateral
Result: Isosceles Triangle

17) Create a class StudentGrade with fields name, marks, and a class variable passingMarks.

Use local variables to assign grade.

```
class StudentGrade {

    // Instance variables
    String name;
    double marks;

    // Class variable
    static double passingMarks = 40;

    // Constructor
    StudentGrade(String name, double marks) {
        this.name = name;
        this.marks = marks;
    }

    // Method to assign grade
    void assignGrade() {

        // Local variable
        String grade;

        if (marks >= 90) {
            grade = "A+";
        } else if (marks >= 80) {
            grade = "A";
        } else if (marks >= 70) {
            grade = "B";
        } else if (marks >= 60) {
            grade = "C";
        } else if (marks >= passingMarks) {
            grade = "D";
        } else {
            grade = "Fail";
        }

        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
        System.out.println("Passing Marks: " + passingMarks);
        System.out.println("Grade: " + grade);
    }
```

```
    public static void main(String[] args) {

        // Local variables
        String studentName = "Sumit";
        double studentMarks = 85;

        StudentGrade s1 = new StudentGrade(studentName, studentMarks);

        s1.assignGrade();
    }
}
```

OUTPUT:
Name: Sumit
Marks: 85.0
Passing Marks: 40.0
Grade: A

18) Create a class ElectricBill with a field units and a class variable fixedCharge. Use local variables to calculate total bill.

```
class ElectricBill {

    // Instance variable
    double units;

    // Class variable
    static double fixedCharge = 100;

    // Constructor
    ElectricBill(double units) {
        this.units = units;
    }

    // Method to calculate total bill
    void calculateBill() {

        // Local variables
        double unitRate = 5;
        double electricityCharge = units * unitRate;
        double totalBill = electricityCharge + fixedCharge;

        System.out.println("Units Consumed: " + units);
```

```
        System.out.println("Fixed Charge: " + fixedCharge);
        System.out.println("Electricity Charge: " + electricityCharge);
        System.out.println("Total Bill: " + totalBill);
    }

    public static void main(String[] args) {

        // Local variable
        double consumedUnits = 200;

        ElectricBill e1 = new ElectricBill(consumedUnits);

        e1.calculateBill();
    }
}
```

OUTPUT:
Units Consumed: 200.0
Fixed Charge: 100.0
Electricity Charge: 1000.0
Total Bill: 1100.0

19) Create a class Product with fields id, name, price, and a class variable discountRate. Use local variables to calculate final price.

```
class Product {

    // Instance variables
    int id;
    String name;
    double price;

    // Class variable
    static double discountRate = 10;

    // Constructor
    Product(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    // Method to calculate final price
```

```
    void calculateFinalPrice() {

        // Local variables
        double discount = (price * discountRate) / 100;
        double finalPrice = price - discount;

        System.out.println("Product ID: " + id);
        System.out.println("Product Name: " + name);
        System.out.println("Original Price: " + price);
        System.out.println("Discount Rate: " + discountRate + "%");
        System.out.println("Discount Amount: " + discount);
        System.out.println("Final Price: " + finalPrice);
    }

    public static void main(String[] args) {

        // Local variables
        int productId = 101;
        String productName = "Laptop";
        double productPrice = 50000;

        Product p1 = new Product(productId, productName, productPrice);

        p1.calculateFinalPrice();
    }
}
```

OUTPUT:
Product ID: 101
Product Name: Laptop
Original Price: 50000.0
Discount Rate: 10.0%
Discount Amount: 5000.0
Final Price: 45000.0

20) Create a class BankAccount with fields balance, and a class variable bankCode. Use local variables to ensure withdrawal does not exceed balance.

```
class BankAccount {

    // Instance variable
    double balance;
```

```
    // Class variable
    static String bankCode = "SBI001";

    // Constructor
    BankAccount(double balance) {
        this.balance = balance;
    }

    // Method to withdraw money
    void withdraw(double amount) {

        // Local variables
        String message;
        double remainingBalance;

        if (amount <= balance) {
            remainingBalance = balance - amount;
            balance = remainingBalance;
            message = "Withdrawal successful.";
        } else {
            message = "Insufficient balance. Withdrawal not possible.";
        }

        System.out.println("Bank Code: " + bankCode);
        System.out.println("Current Balance: " + balance);
        System.out.println("Withdrawal Amount: " + amount);
        System.out.println(message);

        if (amount <= balance + amount) {
            System.out.println("Remaining Balance: " + balance);
        }
    }

    public static void main(String[] args) {

        // Local variables
        double initialBalance = 10000;
        double withdrawAmount = 3000;

        BankAccount account = new BankAccount(initialBalance);

        account.withdraw(withdrawAmount);
    }
}
```

OUTPUT:
Bank Code: SBI001
Current Balance: 7000.0
Withdrawal Amount: 3000.0
Withdrawal successful.
Remaining Balance: 7000.0

Advanced Level (Q21–Q30)
Focus: problem solving, algorithms, object comparisons, static counters, local variables.

21) Create a class Palindrome with a method using local variables to check if a number is palindrome. Add a class variable countChecks.

```
class Palindrome {

    // Class variable
    static int countChecks = 0;

    // Method to check palindrome
    void checkPalindrome(int number) {

        // Local variables
        int originalNumber = number;
        int reverse = 0;

        while (number > 0) {
            int digit = number % 10;
            reverse = reverse * 10 + digit;
            number = number / 10;
        }

        countChecks++;

        if (originalNumber == reverse) {
            System.out.println(originalNumber + " is a Palindrome Number.");
        } else {
            System.out.println(originalNumber + " is not a Palindrome Number.");
        }
    }

    public static void main(String[] args) {

        Palindrome p1 = new Palindrome();
```

```
        p1.checkPalindrome(121);
        p1.checkPalindrome(123);

        System.out.println("Total Checks: " + countChecks);
    }
}
```

OUTPUT:
121 is a Palindrome Number.
123 is not a Palindrome Number.
Total Checks: 2

22) Create a class Armstrong with a method using local variables to check Armstrong number. Add a class variable totalChecks.

```
class Armstrong {

    // Class variable
    static int totalChecks = 0;

    // Method to check Armstrong number
    void checkArmstrong(int number) {

        // Local variables
        int originalNumber = number;
        int sum = 0;
        int temp = number;

        while (temp > 0) {
            int digit = temp % 10;
            sum = sum + (digit * digit * digit);
            temp = temp / 10;
        }

        totalChecks++;

        if (sum == originalNumber) {
            System.out.println(originalNumber + " is an Armstrong Number.");
        } else {
            System.out.println(originalNumber + " is not an Armstrong Number.");
        }
    }
```

```
    public static void main(String[] args) {

        Armstrong a1 = new Armstrong();

        a1.checkArmstrong(153);
        a1.checkArmstrong(123);

        System.out.println("Total Checks: " + totalChecks);
    }
}
```

OUTPUT:
153 is an Armstrong Number.
123 is not an Armstrong Number.
Total Checks: 2

23) Create a class PrimeChecker with a method using local variables to check prime. Add a class variable totalPrimeChecks.

```
class PrimeChecker {

    // Class variable
    static int totalPrimeChecks = 0;

    // Method to check prime number
    void checkPrime(int number) {

        // Local variables
        boolean isPrime = true;

        if (number <= 1) {
            isPrime = false;
        } else {
            for (int i = 2; i <= number / 2; i++) {
                if (number % i == 0) {
                    isPrime = false;
                    break;
                }
            }
        }

        totalPrimeChecks++;
```

```
        if (isPrime) {
            System.out.println(number + " is a Prime Number.");
        } else {
            System.out.println(number + " is not a Prime Number.");
        }
    }

    public static void main(String[] args) {

        PrimeChecker p1 = new PrimeChecker();

        p1.checkPrime(17);
        p1.checkPrime(20);

        System.out.println("Total Prime Checks: " + totalPrimeChecks);
    }
}
```

OUTPUT:
17 is a Prime Number.
20 is not a Prime Number.
Total Prime Checks: 2

24) Create a class Fibonacci with a method to display first N terms using local variables. Add a class variable seriesCount.

```
class Fibonacci {

    // Class variable
    static int seriesCount = 0;

    // Method to display first N terms
    void displayFibonacci(int n) {

        // Local variables
        int first = 0;
        int second = 1;

        System.out.println("Fibonacci Series:");

        for (int i = 1; i <= n; i++) {
            System.out.print(first + " ");
```

```
            int next = first + second;
            first = second;
            second = next;
        }

        seriesCount++;
        System.out.println();
    }

    public static void main(String[] args) {

        Fibonacci f1 = new Fibonacci();

        f1.displayFibonacci(10);

        System.out.println("Total Series Generated: " + seriesCount);
    }
}
```

OUTPUT:
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34

Total Series Generated: 1

25) Create a class Factorial with a method to compute factorial using local variables. Add a class variable totalCalls.

```
class Factorial {

    // Class variable
    static int totalCalls = 0;

    // Method to calculate factorial
    void calculateFactorial(int number) {

        // Local variable
        long factorial = 1;

        for (int i = 1; i <= number; i++) {
            factorial = factorial * i;
        }
```

```
        totalCalls++;

        System.out.println("Factorial of " + number + " = " + factorial);
    }

    public static void main(String[] args) {

        Factorial f1 = new Factorial();

        f1.calculateFactorial(5);
        f1.calculateFactorial(6);

        System.out.println("Total Method Calls: " + totalCalls);
    }
}
```

OUTPUT:
Factorial of 5 = 120
Factorial of 6 = 720
Total Method Calls: 2

26) Create a class Matrix with fields for 2x2 matrix and a class variable matrixType. Use local variables to add and subtract two matrices.

```
class Matrix {

    // Instance variable for 2x2 matrix
    int[][] matrix;

    // Class variable
    static String matrixType = "2x2 Matrix";

    // Constructor
    Matrix(int[][] matrix) {
        this.matrix = matrix;
    }

    // Method to add two matrices
    void add(Matrix m2) {

        // Local variable
        int[][] result = new int[2][2];
```

```
    System.out.println("Addition of Matrices:");

    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            result[i][j] = this.matrix[i][j] + m2.matrix[i][j];
            System.out.print(result[i][j] + " ");
        }
        System.out.println();
    }
}

// Method to subtract two matrices
void subtract(Matrix m2) {

    // Local variable
    int[][] result = new int[2][2];

    System.out.println("Subtraction of Matrices:");

    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            result[i][j] = this.matrix[i][j] - m2.matrix[i][j];
            System.out.print(result[i][j] + " ");
        }
        System.out.println();
    }
}

public static void main(String[] args) {

    // Local variables
    int[][] array1 = {
        {1, 2},
        {3, 4}
    };

    int[][] array2 = {
        {5, 6},
        {7, 8}
    };

    Matrix m1 = new Matrix(array1);
    Matrix m2 = new Matrix(array2);
```

```
        System.out.println("Matrix Type: " + matrixType);

        m1.add(m2);
        m1.subtract(m2);
    }
}
```

OUTPUT:
Matrix Type: 2x2 Matrix

Addition of Matrices:
6 8
10 12

Subtraction of Matrices:
-4 -4
-4 -4

27) Create a class StringReverse with a method using local variables to reverse a string. Add a class variable totalReversals.

```
class StringReverse {

    // Class variable
    static int totalReversals = 0;

    // Method to reverse a string
    void reverseString(String text) {

        // Local variable
        String reversed = "";

        for (int i = text.length() - 1; i >= 0; i--) {
            reversed = reversed + text.charAt(i);
        }

        totalReversals++;

        System.out.println("Original String: " + text);
        System.out.println("Reversed String: " + reversed);
    }
```

```
    public static void main(String[] args) {

        StringReverse s1 = new StringReverse();

        s1.reverseString("Hello");
        s1.reverseString("Java");

        System.out.println("Total Reversals: " + totalReversals);
    }
}
```

OUTPUT:
Original String: Hello
Reversed String: olleH

Original String: Java
Reversed String: avaJ

Total Reversals: 2

28) Create a class WordCounter with a method using local variables to count words in a sentence. Add a class variable language.

```
class WordCounter {

    // Class variable
    static String language = "English";

    // Method to count words
    void countWords(String sentence) {

        // Local variables
        int count = 0;
        boolean inWord = false;

        for (int i = 0; i < sentence.length(); i++) {

            if (sentence.charAt(i) != ' ' && !inWord) {
                count++;
                inWord = true;
            } else if (sentence.charAt(i) == ' ') {
                inWord = false;
            }
```

```
        }

        System.out.println("Language: " + language);
        System.out.println("Sentence: " + sentence);
        System.out.println("Total Words: " + count);
    }

    public static void main(String[] args) {

        WordCounter w1 = new WordCounter();

        // Local variable
        String text = "Java is a powerful programming language";

        w1.countWords(text);
    }
}
```

OUTPUT:
Language: English
Sentence: Java is a powerful programming language
Total Words: 6

29) Create a class Employee with fields name, salary, and a class variable organization. Use local variables to compare salaries of two employees.

```
class Employee {

    // Instance variables
    String name;
    double salary;

    // Class variable
    static String organization = "ABC Company";

    // Constructor
    Employee(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }

    // Method to compare salaries
    void compareSalary(Employee other) {
```

```
        // Local variable
        String result;

        if (this.salary > other.salary) {
            result = this.name + " has a higher salary.";
        } else if (this.salary < other.salary) {
            result = other.name + " has a higher salary.";
        } else {
            result = "Both employees have the same salary.";
        }

        System.out.println("Organization: " + organization);
        System.out.println(this.name + "'s Salary: " + this.salary);
        System.out.println(other.name + "'s Salary: " + other.salary);
        System.out.println(result);
    }

    public static void main(String[] args) {

        // Local variables
        String name1 = "Sumit";
        double salary1 = 50000;

        String name2 = "Rahul";
        double salary2 = 60000;

        Employee e1 = new Employee(name1, salary1);
        Employee e2 = new Employee(name2, salary2);

        e1.compareSalary(e2);
    }
}
```

OUTPUT:
Organization: ABC Company
Sumit's Salary: 50000.0
Rahul's Salary: 60000.0
Rahul has a higher salary.

30) Create a class Library with fields booksAvailable, and a class variable libraryName. Use local variables to update issue and return operations.

```
class Library {
```

```
// Instance variable
int booksAvailable;

// Class variable
static String libraryName = "COER University Library";

// Constructor
Library(int booksAvailable) {
    this.booksAvailable = booksAvailable;
}

// Method to issue a book
void issueBook(int books) {

    // Local variable
    String message;

    if (books <= booksAvailable) {
        booksAvailable = booksAvailable - books;
        message = books + " book(s) issued successfully.";
    } else {
        message = "Books not available for issue.";
    }

    System.out.println(message);
    System.out.println("Books Available: " + booksAvailable);
}

// Method to return a book
void returnBook(int books) {

    // Local variable
    int updatedBooks = booksAvailable + books;

    booksAvailable = updatedBooks;

    System.out.println(books + " book(s) returned successfully.");
    System.out.println("Books Available: " + booksAvailable);
}

// Display library details
void displayDetails() {
    System.out.println("Library Name: " + libraryName);
```

```
        System.out.println("Books Available: " + booksAvailable);
    }

    public static void main(String[] args) {

        // Local variables
        int initialBooks = 10;
        int issueBooks = 3;
        int returnBooks = 2;

        Library l1 = new Library(initialBooks);

        l1.displayDetails();

        System.out.println("\nAfter Issue:");
        l1.issueBook(issueBooks);

        System.out.println("\nAfter Return:");
        l1.returnBook(returnBooks);
    }
}
```

OUTPUT:

```
Library Name: COER University Library
Books Available: 10

After Issue:
3 book(s) issued successfully.
Books Available: 7

After Return:
2 book(s) returned successfully.
Books Available: 9
```